



In The Name of GOD

Genomics Project Report

Genetic Investigation of Sever Developmental and Behavioral Abnormalities in A Consanguineous Family

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Indication for Referral

Investigation of the genetic cause of a severe neurodevelopmental disorder in an individual with parental consanguinity and multiple affected family members.

Clinical Presentation

The proband is a male patient presenting with developmental delay, intellectual disability (non-educable), severe motor impairment, speech disorder, seizures, and self-injurious behavior. His parents are consanguineous. The family also includes two affected sisters, three unaffected brothers, and one unaffected sister.

In the first step, a family pedigree was constructed.

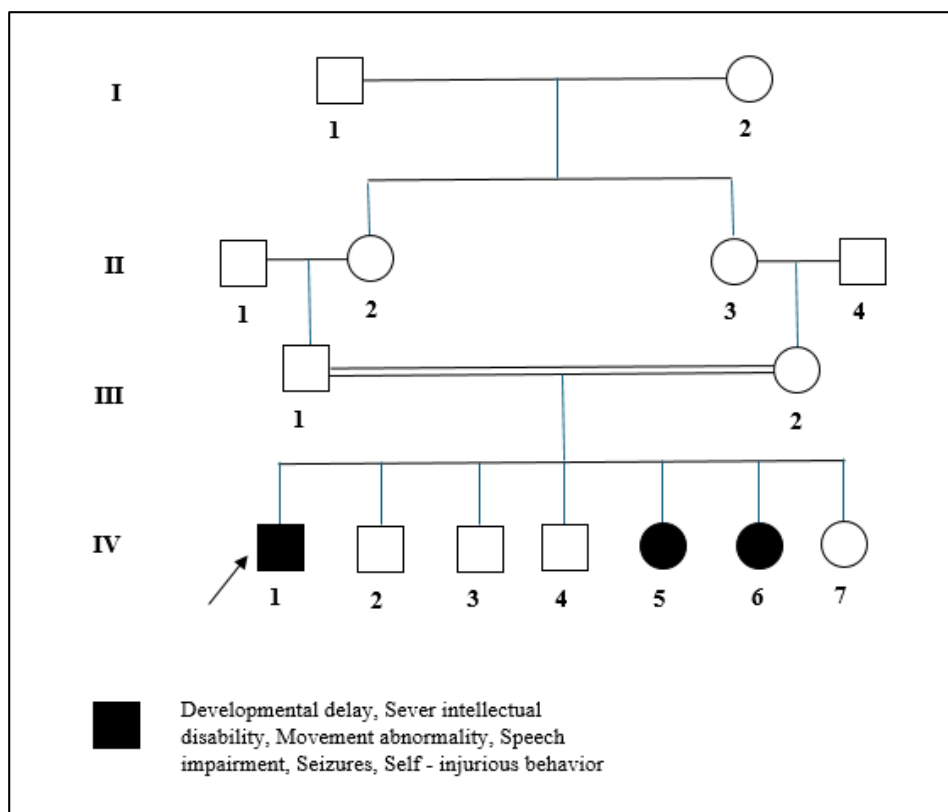


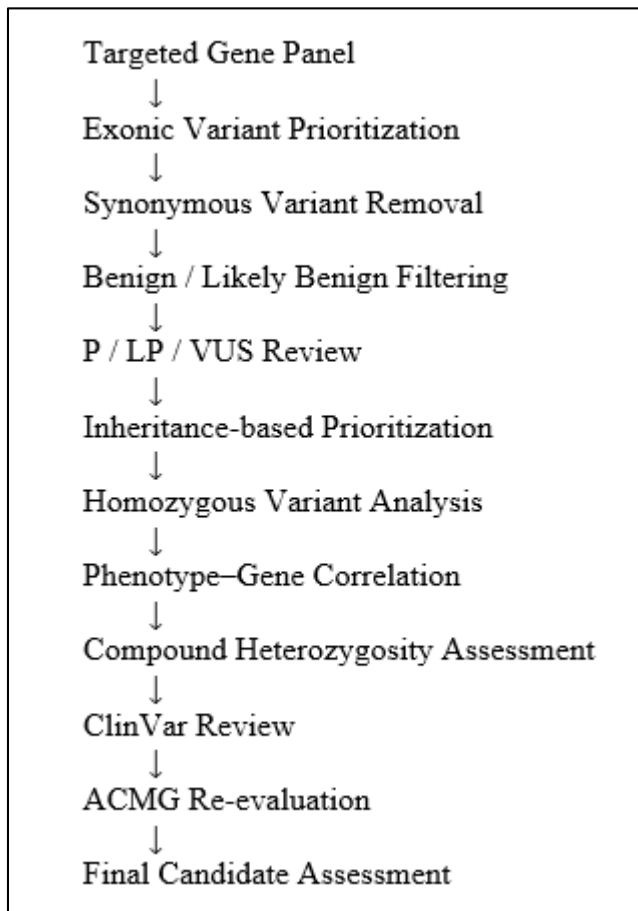
Figure 1. Pedigree of the consanguineous family showing the proband (arrow) and two affected sisters. Parents are first cousins (maternal aunt's daughter and son) and clinically unaffected. Squares represent males, circles represent females; filled symbols indicate affected individuals; open symbols indicate unaffected individuals. The pedigree is consistent with autosomal-recessive inheritance.

Given that the parents are consanguineous (first-cousin marriage) yet clinically unaffected supported an autosomal recessive (AR) inheritance model (suggesting that each is likely a heterozygous carrier of a pathogenic variant), and that both the proband and two sisters are affected (i.e., affected individuals of both sexes within the same sibship), it was hypothesized that the affected individuals may have inherited the pathogenic variant(s) in the homozygous state. Consequently, the initial analysis focused on homozygous variants.

Objective

To identify candidate pathogenic variants and genes potentially responsible for the observed neurodevelopmental phenotype using a targeted gene panel and an inheritance- and phenotype-driven variant prioritization strategy.

Analysis Workflow



Methods

Test type: Targeted gene panel

Filters were applied sequentially as follows:

1. Because the targeted panel focuses on coding regions, analysis prioritized exonic variants; intronic, intergenic, and UTR variants were excluded at this stage.
2. Synonymous variants were filtered out and removed.
3. Based on ClinVar and ACMG classifications, Benign and Likely Benign variants were filtered and excluded from further analysis.
4. Variants classified as Likely Pathogenic, Pathogenic, or VUS were retained for detailed review.

After application of these filters, 25 variants remained and were examined.

Each gene was individually queried in the OMIM database to determine its inheritance pattern and associated clinical phenotypes. Variants within each gene were then cross-checked in ClinVar for current annotations (Pathogenic, Likely Pathogenic, VUS, Benign, Likely Benign). Some variants reported as Pathogenic or Likely Pathogenic in the original annotation file were found, upon direct review in ClinVar, to be currently

classified as VUS, Benign, or Likely Benign. ACMG criteria for these variants were also manually re-evaluated. In several instances, the combination of evidence provided in the original annotation was insufficient to support a Pathogenic or Likely Pathogenic classification. Therefore, in the final interpretation, manual review based on current ClinVar data and ACMG criteria took precedence over the annotations in the Excel file.

Detailed Gene Analysis

Homozygous variants consistent with the pedigree structure were examined first. Genes showing phenotypic overlap were subsequently evaluated for possible compound heterozygosity.

Candidate Homozygous Genes

Genes harboring homozygous variants were prioritized according to the pedigree pattern. Among those reviewed at this stage were TET2, SPG11, and F5. Careful re-evaluation of the variants in these genes showed that they are classified as Benign/Likely Benign in ClinVar. Consequently, none of the homozygous variants provided a convincing explanation for the familial phenotype.

In particular, SPG11 was considered a candidate because of its phenotypic overlap with neurological, motor, and cognitive disorders. According to the literature, SPG11 is associated with hereditary spastic paraplegia type 11(1) and can present with motor impairment, cognitive/developmental deficits, and other neurological manifestations. In the targeted-panel data, a missense variant in SPG11 was identified and reported under alternative transcripts:

Transcript	Variant	Protein change
NM_001160227	c.6730C>T	p.Leu2244Phe
NM_025137	c.7069C>T	p.Leu2357Phe

The variant NM_025137.4: c.7069C>T (p. Leu2357Phe) is a missense change. ClinVar shows conflicting classifications among submitters, with the majority of submissions placing it in the Likely Benign/Benign range. No clear pathogenic intronic or splice-site variant in SPG11 that could constitute a second disease-causing allele was identified in the dataset. Because SPG11-related disease follows autosomal-recessive inheritance, a single missense variant classified as Likely Benign/Benign is insufficient to establish causality.

Genes with Phenotypic Overlap Evaluated for Compound Heterozygosity

In the absence of a convincing homozygous variant, genes whose phenotypes overlap with the patient's presentation were examined. With respect to neurological features, motor impairment, developmental delay, seizures, and speech disorder, the genes PLEKHG5(2), TTN (3), HTT, PAX5 (4), and SPG11 showed varying degrees of phenotypic overlap. Because the observed variants in these genes were predominantly heterozygous, the possibility of two variants within a single gene under an autosomal-recessive inheritance model (compound heterozygosity) was considered. In this scenario, an affected individual would be expected to inherit one pathogenic variant from the father and a second pathogenic variant in the same gene from the mother.

Heterozygous variants were therefore reviewed to determine whether any gene harbored two potentially pathogenic heterozygous alleles. No convincing gene carrying two Pathogenic or Likely Pathogenic

heterozygous variants was identified. The observed variants were classified as Benign or Likely Benign, and no biallelic combination of Pathogenic/Likely Pathogenic alleles was found.

Review of the HTT Gene

HTT shows substantial phenotypic overlap with the patient's neurological and motor features (5,6) and was therefore examined in detail. Seven variants were detected in HTT:

Exon	Variant	Protein change
16	c.2108C>T	p.Pro703Leu
29	c.3801C>T	p.Leu1267Leu
48	c.6591G>A	p.Glu2197Glu
50	c.6925T>C	p.Tyr2309His
52	c.7176A>C	p.Leu2392Leu
57	c.7797A>G	p.Leu2599Leu
61	c.8356G>A	p.Val2786Ile

These variants were reported as synonymous or benign. Among them, the missense variant HTT: NM_002111: c.2108C>T (p. Pro703Leu) is the most notable candidate. This variant is present in ClinVar and has been reported in association with Lopes-Maciél-Rodan syndrome; however, overall classification across submissions is not uniform. In published cases of Lopes-Maciél-Rodan syndrome, p.Pro703Leu has been observed together with a second potentially pathogenic heterozygous variant in HTT.

HTT was not retained as a causal gene for the following reasons:

1. The observed HTT variants were heterozygous and therefore inconsistent with the homozygous inheritance pattern suggested by the family structure.
2. After re-evaluation, the variants were classified as Benign or Likely Benign.
3. A single heterozygous p. Pro703Leu allele is insufficient to establish an HTT-related diagnosis; a second pathogenic allele would be required.
4. Consequently, despite notable phenotypic overlap, the genetic evidence does not support HTT as the cause of disease in this family.

Candidates Selected on the Basis of Phenotypic Similarity

Based on the literature and solely on phenotypic similarity (severe neurodevelopmental disorders, seizures, and motor impairment) together with possible compatibility of the inheritance pattern, the genes WWOX (7), MINPP1(8), AP4B1(9), AP4M1(10), AP4E1(11), AP4S1(12), and HEXB (13) were reviewed. No significant variants were identified in these genes.

De Novo Hypothesis

The possibility that the disease resulted from a de novo variant was also considered. However, the presence of two affected sisters renders an independent de novo event confined to the proband unlikely and poorly compatible with the observed familial pattern; this hypothesis was therefore deprioritized.

Intronic Variant Review

Finally, to ensure completeness, all variants classified as Likely Benign/Benign in the InterVar and CLNSIG columns were filtered out so that any Pathogenic, Likely Pathogenic, or VUS calls located in intronic regions could be examined. No such variants were found.

Recommendations

1. Parental testing is strongly recommended to enable segregation analysis.
2. Orthogonal confirmation of remaining variants of interest (e.g., by Sanger sequencing) is advised.
3. Testing of the affected sisters would further aid interpretation.
4. Re-analysis of the data using updated databases and variant classifications is recommended.
5. The present analysis is limited to a targeted panel. Intronic, regulatory, and structural variants lying outside the panel may have been missed. Whole-exome sequencing (WES) or whole-genome sequencing (WGS) is therefore recommended if clinically feasible.

Conclusion

Given parental consanguinity, the presence of three affected individuals of both sexes, and the severity of the neurological phenotype, a genetic etiology remains highly plausible, and an autosomal-recessive inheritance pattern continues to be a leading hypothesis.

Nevertheless, the current analysis did not identify any homozygous exonic variant that simultaneously:

- is classified as pathogenic according to current ClinVar annotations and ACMG criteria,
- is consistent with the patients' phenotype, and
- matches the family's inheritance pattern.

Evaluation of heterozygous variants likewise failed to reveal a convincing gene carrying two potentially pathogenic alleles. Although genes such as HTT and SPG11 show partial phenotypic overlap with the patient's features, the observed variants lack pathogenic evidence and therefore cannot be supported as causal.

In summary, on the basis of the available data and the analytical scope of the targeted panel, no specific gene or variant can be designated as the cause of disease. This result does not imply a non-genetic etiology; rather, it indicates that a possible genetic cause has not been identified within the limits of the present dataset and analysis.

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